MILITARY SPECIFICATION

HEAD ASSEMBLY, TANK PERISCOPE: 10513392

This amendment forms a part of Military Specification MIL-H-60153(MU), dated 10 February 1965.

Paragraph 3.7.1.1: Delete paragraph and substitute:

* "3.7.1.1 Pre-vibration.- This requirement shall be met prior to meeting the requirements of 3.4. An incident line of sight which is originally parallel to Datum Plane "A" and perpendicular to the intersection of Datum Planes "A" and "B", shown on Drawing F10513392, shall emerge parallel to Datum Plane "B" within 0.20 mil and perpendicular to the intersection of Datum Planes "A" and "B" within 0.60 mil."

Paragraph 3.7.1.2: Delete paragraph and substitute:

* "3.7.1.2 Post-vibration.- This requirement shall be met subsequent to meeting the requirements of 3.4. The incident line of sight specified in 3.7.1.1 shall emerge within 0.10 mil of the collimation readings established in 3.7.1.1."

Paragraph 3.8: Delete paragraph and substitute:

"3.8 Resolution.- Resolution shall be 2.0 seconds of arc or less when observing with a telescope of at least 90 power. The entrance pupil of the telescope shall be not less than 7.0 inches. Resolution shall be observed at one focus setting of the telescope."

Paragraphs 3.9 and 3.10: Delete paragraphs and substitute:

* "3.9 Radiant transmittance.- Minimum luminous transmittance and minimum transmittance at 0.85 micron shall be not less than 90 percent."

Paragraph 4.2.2, Table I: Delete characteristics 112 and 113.

Paragraph 4.3.2: Delete paragraph and substitute the following:

"4.3.2 Environmental.- Three head assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. The samples shall not be selected from those which have previously been vibrated in Table I. They shall be subjected to the following sequence of tests to determine compliance with their respective requirements:



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:12Z
Check the source to verify that this is the current version before use.

| CONTROL TESTS | REQUIREMENTS | TEST PROCEDURES |
|---|---|---|
| 303. Storage temperatures (-80° and +160°F) | 3.3.1 | 4.5.1 |
| 304. Operating temperatures (-40° and +150°F) | 3.3.2 | 4.5.2 |
| 305. Vibration 1/ (60° to 90°F) | 3.4 | 4.5.4" |

1/ Subsequent to being vibrated the head assembly shall be subjected to the tests of 4.5.5 to 4.5.13 inclusive to determine overall compliance with 3.4.

* Paragraph 4.5.8, 4th sentence: Delete and substitute:

"The viewing collimator positioned at the penta prism shall be adjusted for infinity focus and auto-collimated to Datum Plane "A", in elevation, and the intersection of Datum Planes "A" and "B" in azimuth."

* Paragraph 4.5.8, 6th sentence: Delete and substitute:

"The objective lens of the target projector collimator shall be focused for infinity."

* Paragraph 4.5.8, 7th sentence: Delete and substitute:

"Establish the optical axis of the target projector collimator parallel to Datum Plane "A" and perpendicular to the intersection of Datum Planes "A" and "B" by means of auto-collimation. Align the vertical and horizontal reticle lines of the target projector collimator to the vertical and horizontal reticle lines of the viewing collimator."

* Paragraph 4.5.8.4, 5th sentence: After "passing" insert "through".

Paragraph 4.5.9, last sentence: Delete and substitute:

"Resolution shall be within the value specified in 3.8 as determined by observing through the exit aperture of Window 8215997 with auxiliary telescope."

Paragraphs 4.5.10 and 4.5.11: Delete paragraphs and substitute:

* "4.5.10 Radiant transmittance.

4.5.10.1 Luminous transmittance.- Perform this test with the head assembly positioned as in 4.5.9. A diffused luminous source of a color temperature of 2870°K shall be observed with an image forming photometer. The transmittance measurement shall be made



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:12Z
Check the source to verify that this is the current version before use.

while scanning the longitudinal axis of the clear exit window aperture. The transmittance in the scanning area shall meet the value specified in 3.9, measured on the photometer. Compute the percentage of transmittance utilizing the following formula:

$$\frac{\text{Luminance of source}}{\text{Luminance of source as seen through head}} \text{ X100\%}$$

4.5.10.2 Transmittance, 0.85 micron.- Conduct this test in sequence to the test in 4.5.10.1. A diffused light source of approximately 100 foot-lamberts luminance shall be utilized on a target at a distance of 10 feet from the head assembly. The target shall be observed while scanning the longitudinal axis of the clear exit window aperture with the aid of an image forming radiometer. The radiometer shall be filtered for 0.85 micron radiation by a filter which has a peak transmission at 0.85 micron and a bandwidth of ±.01 micron at 1/3 of peak. The transmittance in the scanning area shall meet the value specified in 3.9, as measured on the radiometer. Compute the percentage of transmittance utilizing the following formula:

$$\frac{\text{Filter radiometer reading of target}}{\text{Filter radiometer reading as seen through head"}} \text{ X100\%}$$

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A494Y



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:12Z
Check the source to verify that this is the current version before use.